

Skills Summary

Graphing Parabolas from Vertex or Standard Form

Use this reference while completing your worksheet or when studying.

Skill 1 — Graphing from vertex form

Form: $y = a(x - h)^2 + k$ gives the vertex (h, k) straight away.

$a > 0$ opens upward (vertex is the lowest point); $a < 0$ opens downward (highest point). $|a| > 1$ makes the curve narrower. The x -intercepts come from setting $y = 0$ and taking square roots; the y -intercept comes from putting $x = 0$.

Example: Find the x -intercepts of $y = (x - 3)^2 - 4$.

Step 1. The squared bracket is already alone once the 4 moves across, so square roots are the route — no factoring or formula needed.

Step 2. $(x - 3)^2 = 4$, so $x - 3 = \pm 2$, giving $x = 3 + 2$ and $x = 3 - 2$.

$x = 1$ or $x = 5$, and the vertex $(3, -4)$ sits midway between them.

Try it: Find the x -intercepts of $y = (x + 2)^2 - 9$.

check: $x = 1$ or $x = 5$

(!) **Watch out:** the vertex of $y = (x + 2)^2 - 9$ is at $x = -2$, not $x = 2$. The form is $(x - h)$, so a plus sign inside the bracket means the graph has moved *left*.

Skill 2 — Graphing from standard form

Form: $y = ax^2 + bx + c$.

Axis of symmetry: $x = -\frac{b}{2a}$ — find this first, because the vertex sits on it. **Vertex:** substitute that x back into the equation. **y -intercept:** c . **x -intercepts:** solve $y = 0$ by factoring.

Example: Find the axis of symmetry and the vertex of $y = x^2 - 4x + 3$.

Step 1. The equation is in standard form, so the vertex is not visible — compute the axis first, then substitute, because the vertex always lies on the axis.

Step 2. $a = 1$, $b = -4$, so the axis is $x = -\frac{-4}{2} = 2$; then $y = 2^2 - 4(2) + 3 = -1$.

Axis $x = 2$, vertex $(2, -1)$.

Try it: Find the axis of symmetry and the vertex of $y = x^2 + 6x + 5$.

check: axis $x = -3$, vertex $(-3, -4)$

Skill 3 — Transformations of the parent parabola

Every parabola is the parent graph $y = x^2$ moved about. In $y = a(x - h)^2 + k$: h shifts it sideways (*right* when the bracket says $x - h$), k shifts it up or down, a negative a flips it over the x -axis, and $|a|$ stretches it. Do the sideways move first, then the stretch or flip, then the up-or-down move.

Example: The parent graph is shifted 2 right, flipped over the x -axis, then raised 9. Where does it cross the x -axis?

Step 1. Build the equation one move at a time, then set it to zero — the crossings are the check that the transformations were assembled correctly.

Step 2. The moves give $y = -(x - 2)^2 + 9$; setting $y = 0$ gives $(x - 2)^2 = 9$, so $x - 2 = \pm 3$.
 $x = -1$ or $x = 5$

Try it: The parent graph is shifted 1 left and 16 down. Where does it cross the x -axis?

$x = x$ to $x = x$ check